22AIE312 BIG-DATA ANALATICS

REAL-TIME AIR QUALITY MONITORING AND POLLUTION PREDICTION

Group 2

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Introduction

Domain:

- Environmental Science, Big Data Analytics.

Context:

- Air pollution is a major global health crisis. With the rise of Smart Cities, millions of IoT sensors are generating massive streams of environmental data (Volume & Velocity).

Importance:

- Real-time monitoring is crucial for public health warnings.
- Predictive analytics can help governments take preemptive measures to control AQI (Air Quality Index) levels.
- Traditional data processing tools cannot handle the scale of historical and real-time sensor data effectively.

Problem Statement

To develop a scalable Big Data framework that ingests, processes, and analyzes large-scale air quality data to accurately predict future pollution levels using distributed computing techniques.

5v's :

Dimension	Project Implementation
1. Volume	• 4GB of dataset (zipped file)
2. Velocity	• In-Memory Computing: Using Apache Spark to process data faster than disk-based engines.
3. Veracity	• Data Cleaning: Using Spark RDDs & DataFrames to remove noise and null values from raw sensor inputs. • Reliability: Spark RDD lineage ensures zero data loss during processing failures.
4. Value	• Public Health: Generating real-time warnings for hazardous air quality . • Policy: Enabling predictive analytics for preemptive pollution control measures.
5.Variety	• multiple pollutant concentrations (PM2.5, PM10, NO2, SO2, CO, O3) and environmental parameters like temperature, humidity, and wind speed.

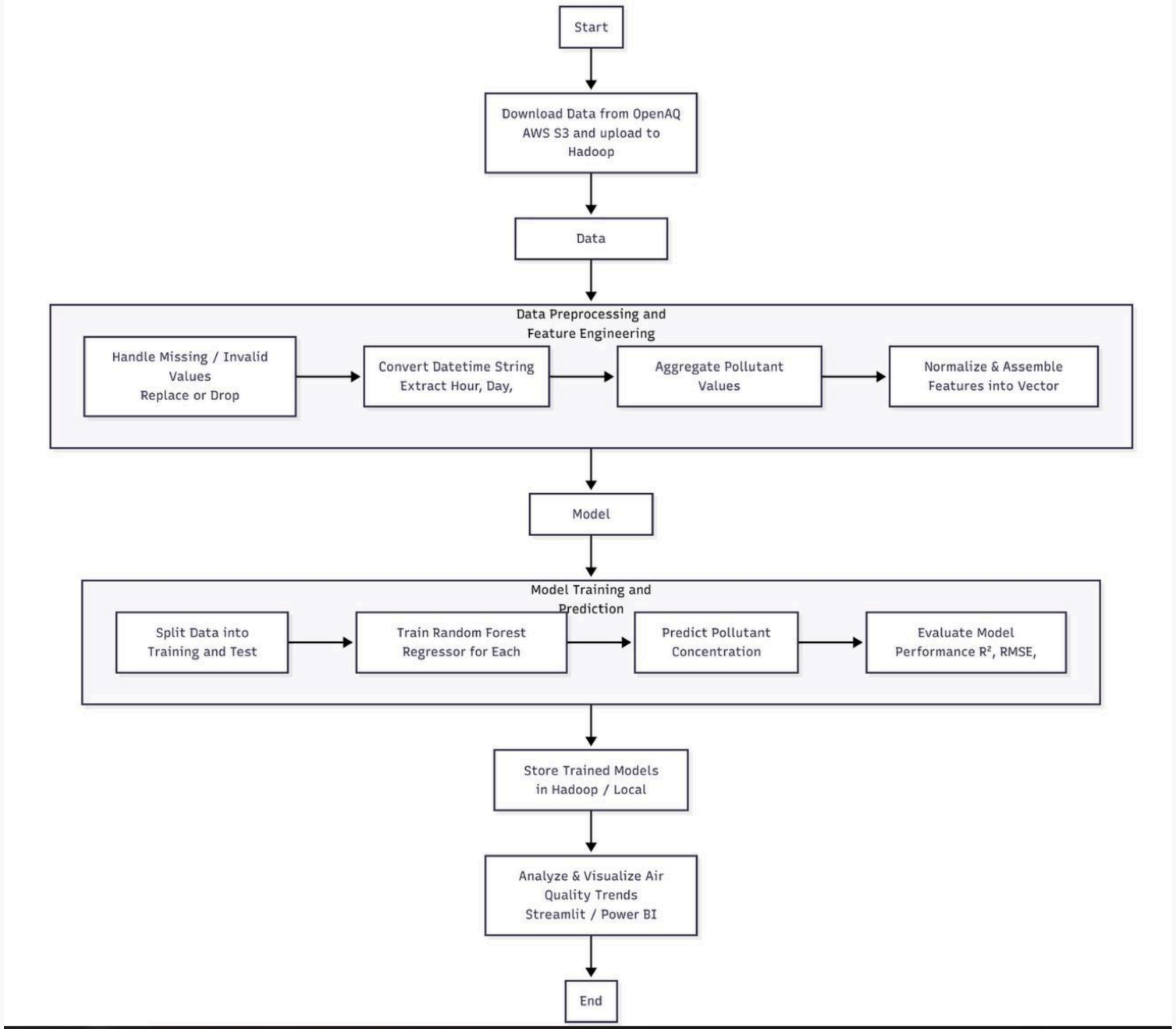
Objectives

- To design a scalable Big Data architecture for storing and managing large volumes of air quality data using MongoDB/HDFS.
- To implement data preprocessing and feature extraction using Apache Spark (RDDs and DataFrames).
- To evaluate the performance of distributed machine learning models (via Spark MLlib) in predicting AQI levels.
- To visualize the pollution trends and prediction results for better decision-making.

Literature Review

S. No.	Topic	What Literature Says	Technologies / Tools Used
1	Spark-Based Prediction Frameworks	• Proposed a distributed framework handling high-velocity PM2.5 data streams. • Proved that Ensemble models (combining multiple algorithms) on Spark outperform single models (R^2 up to 0.76) for short-term forecasting	• Apache Spark (Big Data Framework) • Spark MLlib • Ensemble Learning (Random Forest, GBDT)
2	Comparative Analysis of ML Models	• Compared Decision Tree, Random Forest (RF), and ANN for AQI prediction. • Found that Random Forest offered the best balance of accuracy and processing speed when scaled on a cluster, outperforming Deep Learning in training time.	• Distributed Computing (Cloud) • Random Forest Regression • Decision Tree Regression
3	Big Data vs. Traditional ML	• Demonstrated that while traditional ML (e.g., Scikit-Learn) works for small datasets, it fails with high-frequency sensor logs. • Validated that PySpark significantly reduces latency for cleaning and aggregating multi-city air quality datasets.	• PySpark (Python + Spark) • HDFS (Storage) • Matplotlib (Visualization)
4	IoT & Real-Time Monitoring	• Highlighted that real-time monitoring requires a layered architecture. • Emphasized data preprocessing (null removal, normalization) as the most critical step for accuracy.	• IoT Sensors (nodeMCU/ESP8266) • Cloud Platforms • Pre-processing Algorithms

Proposed System



Implementation Details

Technologies Used:

- Apache Spark, Spark MLlib, Hadoop HDFS, AWS CLI, JVM 11+, Scala

Software Architecture:

- Parallel data ingestion and preprocessing using Spark DataFrames and RDDs
- Distributed model training using Spark MLlib for pollutant prediction
- Models and results stored in Hadoop (HDFS) for scalability and fault tolerance
- End-to-end pipeline integrates data collection, cleaning, feature engineering, model training, and evaluation

Hardware / Software Requirements:

1. **Operating System:** Linux /Windows with WSL (Ubuntu).
2. **Storage:** HDFS .
3. **Processing Engine:** Apache Spark .
4. **Programming Language:** Scala and Python (PySpark).
5. **Machine Learning:** Spark MLlib

Datasets:

- https://zenodo.org/records/17085064

Base Paper:

- https://www.mdpi.com/1660-4601/18/13/7087

Dataset Overview

- **Dataset:** Air Pollution in China (2015–2025)
- **Source:** Kaggle Dataset Repository
- **Records:** Large-scale time-series air pollution data collected from multiple monitoring stations across China covering 2015–2025.
- **Attributes:**
  - PM2.5 – Fine particulate matter concentration
  - PM10 – Coarse particulate matter concentration
  - NO_2 – Nitrogen Dioxide level
  - SO_2 – Sulfur Dioxide level
  - CO – Carbon Monoxide concentration
  - O_3 – Ozone level
- Temperature – Ambient temperature readings
- Humidity – Atmospheric humidity level
- Wind Speed – Wind movement affecting pollution dispersion
- Date & Time (Timestamp) – Time-based sensor measurement records

Dataset Ingestion into Hadoop (HDFS)

- A large-scale air quality dataset (4 GB, Air Pollution in China 2015–2025) containing pollutant concentrations and environmental parameters was used.
- The dataset was uploaded into the Hadoop Distributed File System (HDFS) to enable distributed storage and parallel processing.
- Local storage was not used for analysis; HDFS acted as the primary data source for all processing tasks.
- The dataset consists of time-series sensor data, making it suitable for big data analytics and prediction models.
- Raw air quality data was stored in:
  - /datasets/BDA1
- HDFS commands were used to verify:
  - Successful data upload
  - File accessibility across the cluster

RESULTS

Hadoop | Overview | Datanodes | Datanode Volume Failures | Snapshot | Startup Progress | Utilities

Browse Directory

/BDA1 Go!

Show 25 entries Search:

Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name
drwxr-xr-x	spark	supergroup	0 B	Mar 07 11:04	0	0 B	Models
-rwxrwxrwx	spark	supergroup	3.58 GB	Feb 23 10:25	3	128 MB	china-AQ-dataset.csv
drwxr-xr-x	spark	supergroup	0 B	Feb 23 10:51	0	0 B	processed_china_AQ

Showing 1 to 3 of 3 entries

Previous 1 Next

Hadoop, 2019.

RESULTS

Hadoop Overview Datanodes Datanode Volume Failures Snapshot Startup Progress Utilities

Browse Directory

/BDA1/processed_china_AQ Go!

Show 25 entries Search:

	Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name
	-rw-r--r--	spark	supergroup	0 B	Feb 23 10:51	3	128 MB	_SUCCESS
	-rw-r--r--	spark	supergroup	2.19 GB	Feb 23 10:51	3	128 MB	part-00000-8e1d54ce-dfec-4a46-b098-25d560c01c1d-c000.csv

Showing 1 to 2 of 2 entries Previous 1 Next

Hadoop, 2019.

RESULTS

Hadoop | Overview | Datanodes | Datanode Volume Failures | Snapshot | Startup Progress | Utilities

Browse Directory

/BDA1/Models Go!

Show 25 entries Search:

Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name
drwxr-xr-x	spark	supergroup	0 B	Feb 23 13:03	0	0 B	co_model
drwxr-xr-x	spark	supergroup	0 B	Feb 23 13:20	0	0 B	no2_model
drwxr-xr-x	spark	supergroup	0 B	Feb 23 12:45	0	0 B	o3_model
drwxr-xr-x	spark	supergroup	0 B	Mar 07 11:04	0	0 B	pm10_model
drwxr-xr-x	spark	supergroup	0 B	Feb 23 12:31	0	0 B	pm25_model
drwxr-xr-x	spark	supergroup	0 B	Feb 23 13:39	0	0 B	so2_model

Showing 1 to 6 of 6 entries

Previous 1 Next

Hadoop, 2019.

RESULTS

Spark Master at spark://172.20.0.6:7077

URL: spark://172.20.0.6:7077
Alive Workers: 1
Cores in use: 20 Total, 0 Used
Memory in use: 6.6 GiB Total, 0.0 B Used
Resources in use:
Applications: 0 Running, 1 Completed
Drivers: 0 Running, 0 Completed
Status: ALIVE

Workers (1)

Worker Id	Address	State	Cores	Memory	Resources
worker-20260223063448-172.20.0.10-42929	172.20.0.10:42929	ALIVE	20 (0 Used)	6.6 GiB (0.0 B Used)	

Running Applications (0)

Application ID	Name	Cores	Memory per Executor	Resources Per Executor	Submitted Time	User	State	Duration

Completed Applications (1)

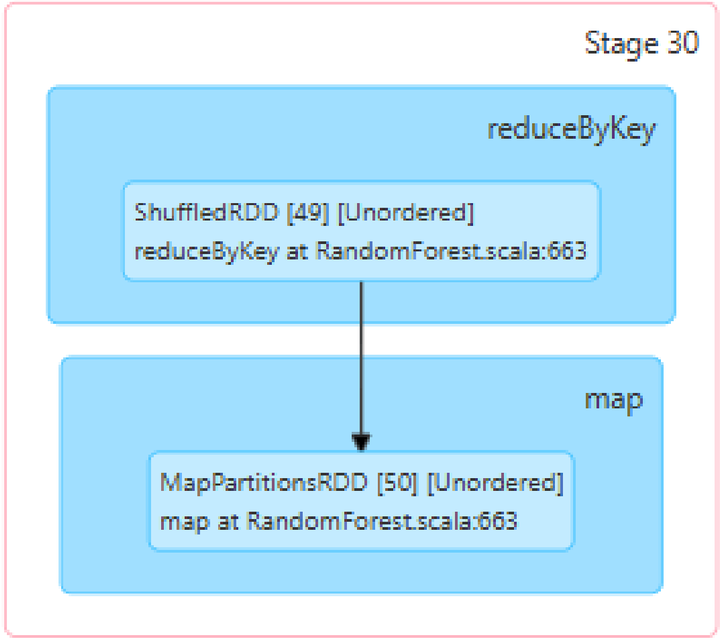
Application ID	Name	Cores	Memory per Executor	Resources Per Executor	Submitted Time	User	State	Duration
app-20260223063704-0000	AirQualityModel_Training	20	4.0 GiB		2026/02/23 06:37:04	spark	FINISHED	1.5 h

RESULTS

Details for Stage 30 (Attempt 0)

Resource Profile Id: 0
Total Time Across All Tasks: 2 s
Locality Level Summary: Node local: 21
Shuffle Read Size / Records: 166.7 MiB / 134400
Associated Job Ids: 12

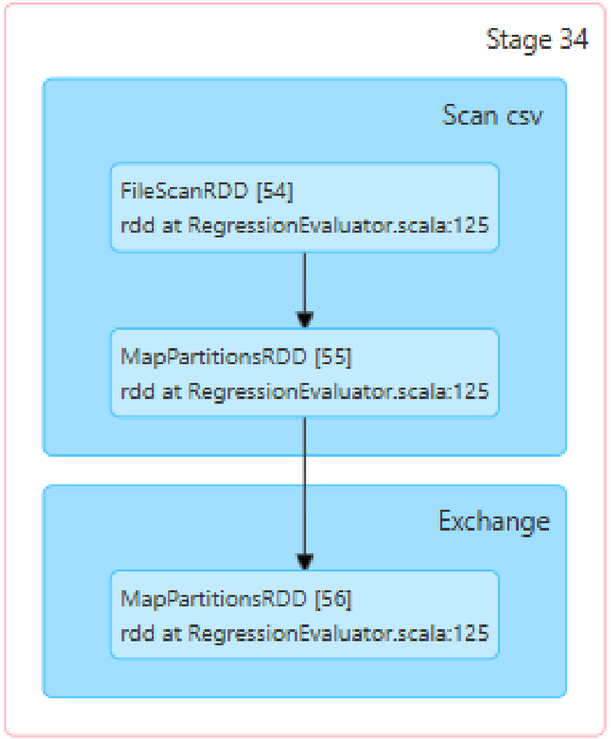
▾ DAG Visualization



Details for Stage 34 (Attempt 0)

Resource Profile Id: 0
Total Time Across All Tasks: 4.0 min
Locality Level Summary: Any: 20
Input Size / Records: 2.2 GiB / 31984197
Shuffle Write Size / Records: 934.6 MiB / 31984197
Associated Job Ids: 14

▾ DAG Visualization



RESULTS

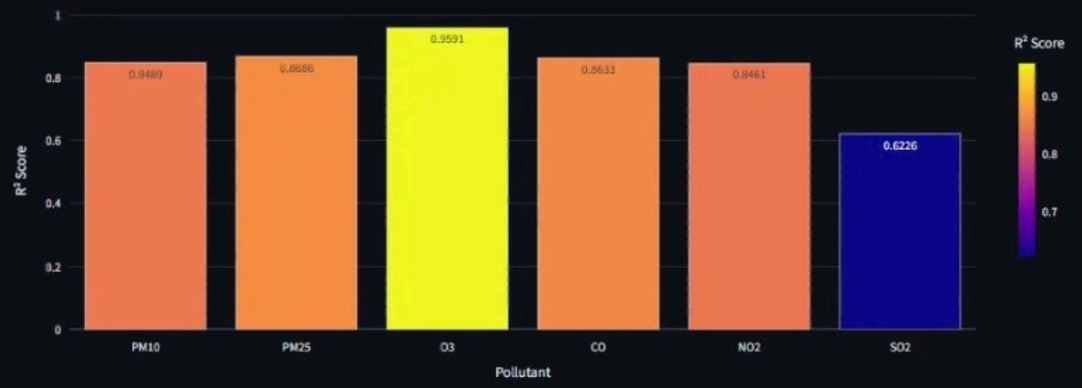
Air Quality Prediction System

Distributed Big Data & Machine Learning (Spark + HDFS + FastAPI)

Model Performance | Live Prediction

Model Accuracy Comparison

	Pollutant	R^2 Score
0	PM10	0.8489
1	PM25	0.8686
2	O3	0.9591
3	CO	0.8633
4	NO2	0.8461
5	SO2	0.6226



PM10	PM25	O3	CO	NO2	SO2
0.8489	0.8686	0.9591	0.8633	0.8461	0.6226

Built with Apache Spark • Hadoop HDFS • FastAPI • Streamlit

Challenges / Limitations

- Faced difficulty finding a reliable and structured data source for air quality data.
- The OpenAQ API had access and response issues, preventing live data retrieval for dashboard integration.
- The historical data stored in AWS S3 was not properly structured, making extraction complex.
- Faced challenges in downloading India's dataset due to inconsistent file organization.
- As a result, downloaded and used the China dataset folder, which was more accessible and complete.

Impact Comparison Between Big Data Tools and Conventional Data Processing Techniques

- **Conventional Techniques:** Slow, sequential, not suited for dynamic/large datasets; limit experimentation and real-time deployment.
- **Big Data Tools (e.g., Apache Spark):** Enable parallel processing, efficient data pipeline scaling, and agile iteration over massive datasets. Essential for building next-generation, real-time, and scalable clinical decision support solutions. Demonstrated near-linear speedup and robust, fault-tolerant data ingestion

Conclusion & Future Work

- Utilized large-scale air quality datasets and Apache Spark for distributed data processing.
- Applied Spark MLlib Random Forest models for pollutant prediction and environmental data analysis.
- Demonstrated the effectiveness of Big Data and Machine Learning in handling and predicting air quality trends.
- The system provides scalable data processing and visualization through an interactive dashboard.
- Includes integrating real-time data streaming, advanced ML models, and IoT-based air quality monitoring systems.

References

- Kang, Gaganjot Kaur, Jerry Zeyu Gao, Sen Chiao, Shengqiang Lu, and Gang Xie. "Air quality prediction: Big data and machine learning approaches." Int. J. Environ. Sci. Dev 9, no. 1 (2018): 8-16.
- Bosco, J. John, and V. Kowsalya. "A Novel Approach for Air Pollution Prediction Using Machine Learning Techniques." In 2024 Third International Conference on Electrical, Electronics, Information and Communication Technologies (ICEEICT), pp. 1-6. IEEE, 2024.
- Kothandaraman, D., N. Praveena, K. Varadarajkumar, B. Madhav Rao, Dharmesh Dhabliya, Shivaprasad Satla, and Worku Abera. "Intelligent forecasting of air quality and pollution prediction using machine learning." Adsorption Science & Technology 2022 (2022): 5086622.

Thank you